

1. Write a program to check whether a number is a Strong number or not.

A Strong number is a special number whose sum of factorial of digits is equal to the original number.

For example: 145 is a strong number, since

$$1! + 4! + 5! = 145$$

[Hint: conditional operator, method, use parameterized method to take input]

Sample Input 1:

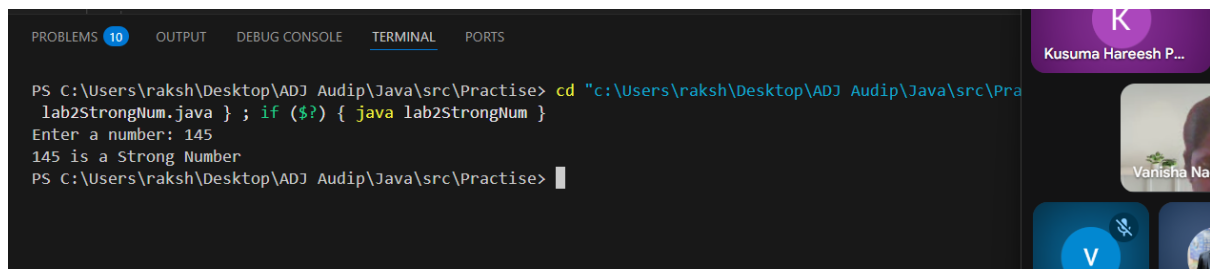
145

Expected Output:

145 is a Strong Number

```
J lab2StrongNum.java
1  import java.util.Scanner;
2
3  public class lab2StrongNum {
4
5      static void checkStrong(int num) {
6          int original = num;
7          int sum = 0;
8          while (num > 0) {
9              int digit = num % 10;
10             int fact = 1;
11             for (int i = 1; i <= digit; i++) {
12                 fact *= i;
13             }
14             sum += fact;
15             num /= 10;
16         }
17
18         String result = (sum == original)
19             ? original + " is a Strong Number"
20             : original + " is NOT a Strong Number";
21
22         System.out.println(result);
23     }
24
25     public static void main(String[] args) {
26         Scanner sc = new Scanner(System.in);
27
28         System.out.print("Enter a number: ");
29         int number = sc.nextInt();
30         checkStrong(number);
31         sc.close();
32     }
33 }
34
```

OUTPUT:



The image shows a terminal window with a dark background. The terminal output is as follows:

```
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise> cd "c:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise"
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise> java lab2StrongNum.java ; if ($?) { java lab2StrongNum }
Enter a number: 145
145 is a Strong Number
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise>
```

On the right side of the terminal window, there is a Zoom meeting sidebar. It shows three participants: 'Kusuma Hareesh P...' with a purple profile picture, 'Vanisha Na...' with a white profile picture, and 'V' with a blue profile picture.

2. Write a program to input two numbers and find the maximum between the two numbers using the conditional/ternary operator.

Sample Input:

num1 = 10

num2 = 30

Expected Output:

The maximum between 10 and 30 is: 30

```
J labMaxNumber.java > labMaxNumberber
1  import java.util.Scanner;
2
3  public class labMaxNumber {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          System.out.print("Enter num1: ");
8          int num1 = sc.nextInt();
9
10         System.out.print("Enter num2: ");
11         int num2 = sc.nextInt();
12         int max = (num1 > num2) ? num1 : num2;
13
14         System.out.println("The maximum between " + num1 + " and " + num2 + " is: " + max);
15
16         sc.close();
17     }
18 }
19
```

OUTPUT:

```
PROBLEMS 10 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise> cd "c:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise" & if ($?) { java lab2StrongNum }
Enter a number: 145
145 is a Strong Number
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise> cd "c:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise" & if ($?) { java labMaxNumber }
Enter num1: 10
Enter num2: 30
The maximum between 10 and 30 is: 30
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise>
```